

Jiashu Liu

Electrical Engineering & Automation Student | Embedded Systems, Robotics and Industrial Automation

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Education

Chongqing University - University of Cincinnati Joint Co-op Institute 2022 - 2027
B.Eng. in Electrical Engineering and Automation, expected June 2027

Skills

Embedded & IoT	STM32, C51, Tuya MCU SDK, UART, Bluetooth/Wi-Fi adaptation, reusable templates
Hardware Debugging	PCB bring-up, oscilloscope diagnosis, power-ripple reduction, ESC main-control/current-board debugging
Automation & Robotics	Robot workflow design, point calibration, IO/register use, pneumatics, commissioning
Delivery	Requirement breakdown, test evidence, WI writing, operator training and production handover

Co-op Experience

Beijing Siemens Cerberus Electronics Limited | Automation Engineering Intern Jan 2025 - Apr 2025

- Supported AD40i O/H/S board changeover, soldering-point verification and program debugging for multiple line configurations.
- Programmed and optimized an Aubo oil-injection robot station, including TCP communication, IO/register use, cylinder control, point setting and aging verification.
- Improved the oiling process with a quantitative valve and limit structure: 1,050 small-valve trial units with 0 failures; improved large-valve trial success about 95.4%.
- Wrote WI documents and trained production-line staff for repeatable handover.

Beijing Siemens Cerberus Electronics Limited | Automation Engineering Intern May 2024 - Aug 2024

- Worked on AD40i/QFM2 semi-automatic soldering equipment, S-board soldering program writing, point calibration and production issue support.
- Joined the TEVA small-part robot station build from gripper selection and wiring to program writing, helping move the equipment into trial production.

Engineering Projects

Standardized IoT Access Solution for Motor Control Systems | Main Developer Sep 2025 - Dec 2025

- Built a wireless debugging and IoT access solution using Tuya WBR3, BT7L-IPEX, STM32 UART parsing and SDK adaptation.
- Used an oscilloscope to locate USB-to-TTL power ripple issues and reduced 3.3 V ripple from about 150 mV to within 30 mV.
- Refactored the Tuya MCU SDK with HAL Adapter, UART_IDLE interrupt, ring buffer and event-driven DP model; new devices can be connected in about one hour with the template.

2025 Smart Car Competition | Hardware and MicroPython Contributor 2025

- Worked on micro-photoelectric and speed-photoelectric cars: ESC main-control/current boards, vehicle-model design, mainboard work and phototube-module soldering/debugging.
- Independently wrote the MicroPython program for the speed-photoelectric group; team won first prize in the school competition and was selected for the western regional competition.

C51 Line-Tracking and Obstacle-Avoidance Vehicle | Team Leader / IR Remote Module Jan 2024 - Apr 2024

- Led a four-person STC89C52RC / C51 coursework prototype; used Keil C51 and Proteus for simulation and debugging.

Research and Awards

Sensor-Centric Perception for Unmanned Surface Vehicles | Submitted | First Author

Review topic: USV perception systems, datasets, multi-modal sensor fusion and edge intelligence; approximately 6,887 words with 10 figures and 4 tables.

- 2025 First Prize, Chongqing University Smart Car Competition
- 2024 Second Prize, Chongqing University Mathematical Contest in Modeling / MCM-related competition